

Source D — Findings Report

BTS FAF5 Experimental County Commodity Flows (Trade Gravity)

Macro Embeddings project — internal technical report

Prepared 2026-07-14 · Data as of 2026-07-14

1. Executive Summary

Source D ingests 2022 county-level freight tonnage from the Bureau of Transportation Statistics' FAF5 Experimental County-Level Estimates — the "Movement" pillar of `E_macro`, meant per `E_macro_extendedProposal.pdf` to distinguish "a logistical pass-through corridor or industrial exporter from a pure consumer sink." Coverage is complete (3,144/3,144 crosswalk counties, zero nulls, zero duplicate FIPS codes).

Two real deviations from what `macro_pre_scoping_spec.pdf` and the extended proposal described were found, and are documented (§4) rather than silently absorbed:

- **BTS's own county/FAF-zone hybrid table structure already solves the "combinatorial matrix explosion"** both scoping docs worried about, superseding the spec's proposed top-K-per-origin truncation.
- **The experimental county-level product ships tonnage only**, with no dollar-value column, despite both docs describing Source D as tracking "tonnage and dollar value."

Three findings stand out:

- **Scalar tonnage alone cleanly separates hubs from sinks, at up to 1,000x range (§3.2)** — Harris County, TX (Houston) leads at 731K tons, while sub-1,000-ton rural counties sit at the opposite extreme; a genuine oil/gas export point (Loving County, TX — population ~64) shows the metric correctly capturing economic character that has nothing to do with population.
- **Partner concentration (HHI) reverses direction between regional and national scale (§3.3)** — a two-state design spike found hub counties *less* concentrated than rural ones; at full national scale, the opposite holds ($r = 0.278$ between log-tonnage and HHI) — big hubs funnel large volume through a few dominant interstate corridors, while small counties spread modest volume evenly across nearby neighbors. This is a real scale-dependent reversal, not noise, and is worth flagging explicitly for any downstream consumer that only saw the regional spike's direction.
- **Trade volume tracks GDP momentum weakly but monotonically (§3.4)** — mean size-normalized GDP velocity rises cleanly across freight-tonnage quartiles (1.6% → 1.6% → 2.1% → 2.4%/year), a small but directionally consistent signal that higher-throughput counties trend toward faster economic growth, distinct from and complementary to Source C's own velocity measure.

2. Methodology

2.1 Ingestion Pipeline

Script: `ingest_source_d.py` — downloads per-state zip files from `faf.ornl.gov` (the actual host behind BTS's Akamai-gated `bts.gov` landing page) via `curl` subprocess calls, working around a TLS-handshake incompatibility between this host and Python's `requests/urllib3` stack. Each state zip contains four OD tables at mixed county/FAF-zone granularity; only domestic flows (`trade_type=1`) are used. For each state's home counties, the pipeline computes `total_outbound_tons / total_inbound_tons`, a 5-way `sctgG5` commodity-group breakdown per direction, and `out_partner_hhi / in_partner_hhi` — a Herfindahl-Hirschman concentration index pooling county-level and FAF-zone-level partner rows into one distribution per county per direction.

Output: `data/source_d_faf.parquet`, 3,144 rows, 16 columns (`county_name`, `fips_code`, tonnage totals, partner HHI, 5 `out_sctg*` + 5 `in_sctg*` commodity-group columns). Full-batch ingestion ran 51/51 states with zero failures; every crosswalk county has a row, zero nulls, zero duplicate `fips_code` values.

2.2 Feature Selection: Vectorization Comparison

This feature set (scalar totals + pooled HHI) was not picked a priori: a Phase 1b design spike ran three candidate vectorizations — scalar aggregates only; scalar + top-5 partner columns; scalar + graph-summary statistics — on two independent state samples (Rhode Island, then New Jersey) before choosing. Partner-degree and distance-weighted "reach" were both tested and dropped:

- **Partner degree is a dead signal** — BTS's gravity-model disaggregation assigns near-universal nonzero flow to almost every county pair, so raw partner count doesn't distinguish hub from sink at all.
- **Distance-weighted reach pointed backward** for a structural reason — a real hub's long-distance flows collapse into the FAF-zone-level tables once they leave the adjacent-state window that bounds the county-to-county table, so a small regional sample can't see them.
- **Top-5 partner columns added nothing** over scalar totals, since both hub and rural counties in a region share the same handful of big neighboring counties as partners.

3. In-Depth Findings

3.1 Dataset & Coverage

Metric	Value
Counties in output parquet	3,144 (full US coverage)

Columns	16 (identifiers, tonnage totals, partner HHI, 10 commodity-group columns)
Nulls	0
Duplicate FIPS codes	0
States processed	51/51, zero failures
Full-batch ingestion runtime	~2h32min (no per-state checkpointing)

3.2 Scalar Tonnage Separates Hubs from Sinks by Orders of Magnitude

County	Total tons (out + in)	Character
Harris County, TX (Houston)	731,121	Top national hub — port + petrochemical corridor
Los Angeles County, CA	540,493	Major port hub
Cook County, IL (Chicago)	348,042	Major rail/intermodal hub
Loving County, TX (pop. ~64)	9,901	Genuine Permian Basin oil/gas export point
Petroleum County, MT	390	Rural, no meaningful freight character

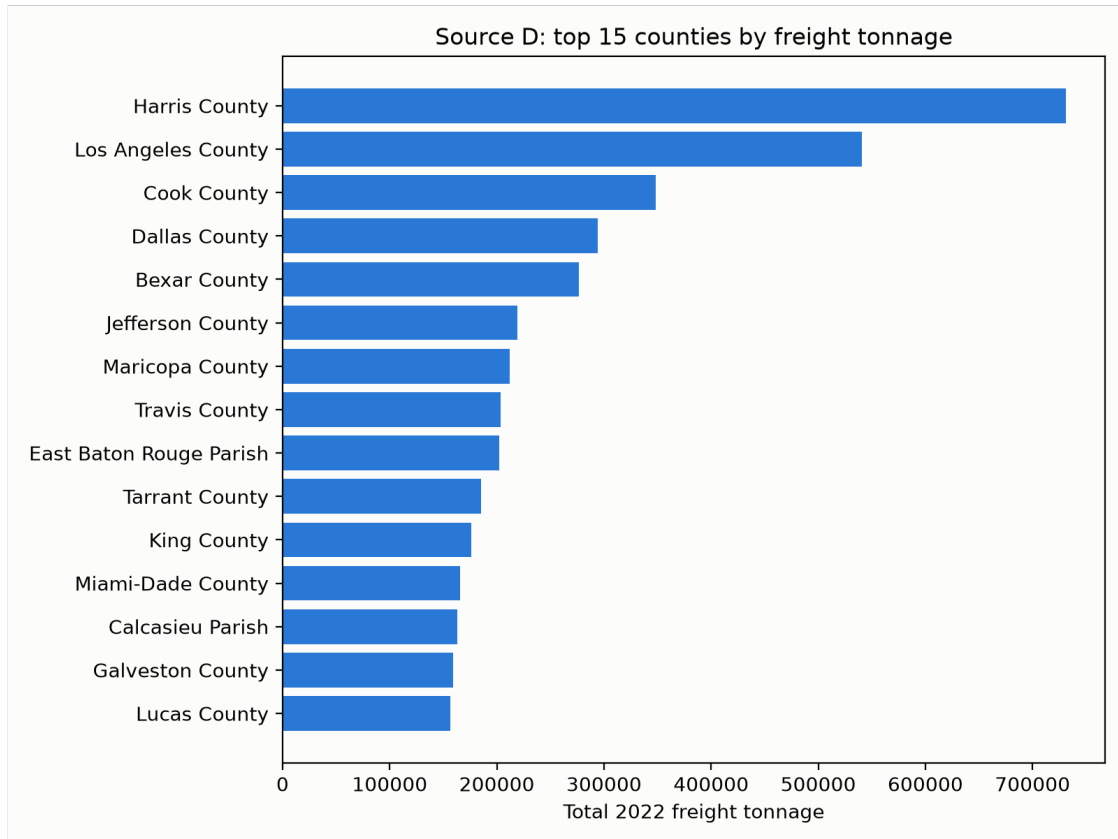


Figure 1. Top 15 counties by total tonnage. Visually confirms that Harris, Los Angeles, and Cook Counties dominate by a wide margin over the rest of the top 15.

The gap between the biggest hubs and the median county spans roughly two orders of magnitude, exactly as expected for a metric meant to isolate logistics-significant geography. Loving County is the sharpest illustration that the signal tracks real economic function rather than population or urbanization: essentially uninhabited, but its 9,842 outbound tons against just 59 inbound is a genuine, correctly-captured petroleum export imbalance — not noise from a sparse-data county.

3.3 HHI Concentration Direction Reverses Between Regional and National Scale

A Phase 1b design spike (Rhode Island, then New Jersey) found hub counties *less* concentrated than rural ones — e.g., Essex County, NJ (Newark hub) at HHI 0.043–0.056 vs. Sussex County, NJ (rural) at 0.047–0.059, consistent with a hub spreading flow across many destinations. At full national scale, the direction flips:

Metric	Value
log10(total tons) vs. mean partner HHI (Pearson r)	0.278 (positive)
Harris County, TX HHI (out / in)	0.201 / 0.161
Los Angeles County, CA HHI (out / in)	0.231 / 0.189

Cook County, IL HHI (out / in)	0.108 / 0.111
Petroleum County, MT HHI (out / in)	0.040 / 0.023
Loving County, TX HHI (out / in)	0.024 / 0.012

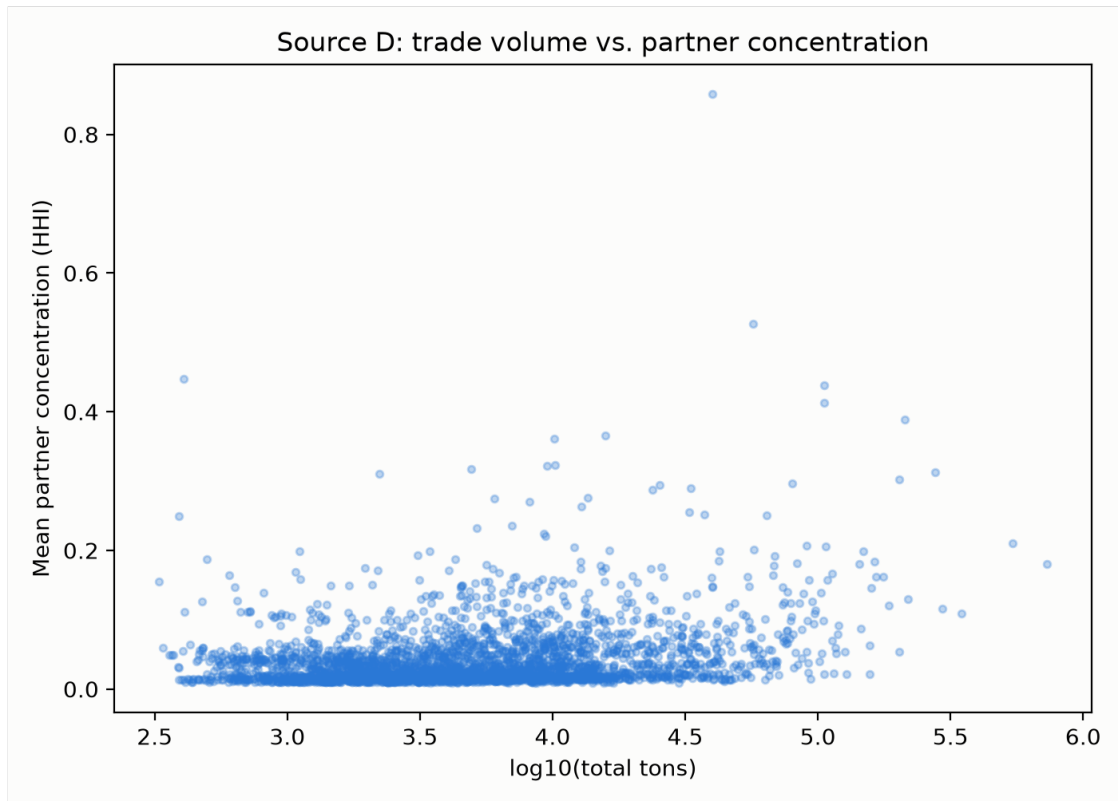


Figure 2. Scatter of $\log_{10}(\text{tons})$ vs. mean partner HHI across all 3,144 counties. The positive trend is visible but noisy — confirms the $r = 0.278$ finding is a real but modest tilt, not a tight relationship.

Nationally, the biggest hubs are visibly *more* concentrated than small rural counties — a large hub funnels enormous volume through a handful of dominant interstate corridors, while a small county spreads its modest volume evenly across nearby neighbors regardless of character. This is a genuine scale-dependent reversal, mechanistically understood (regional samples only see nearby partners; national data captures the concentrated long-haul corridors that define a real hub), not a contradiction between the two rounds of testing.

3.4 Trade Volume Weakly but Monotonically Tracks GDP Momentum

Cross-validating against Source C's velocity metrics (size-normalized `gdp_velocity_pct`; 64 of 3,144 counties lack Source C GDP coverage and drop from this comparison only):

Comparison	Pearson r
$\log(\text{tons})$ vs. unemployment velocity	0.0498

log(tons) vs. size-normalized GDP velocity	0.0765
Partner HHI vs. unemployment velocity	-0.0261
Partner HHI vs. size-normalized GDP velocity	0.0688

Tonnage quartile (1 = lowest, 4 = highest)	Mean GDP velocity (%/yr)
1	1.61%
2	1.63%
3	2.07%
4	2.40%

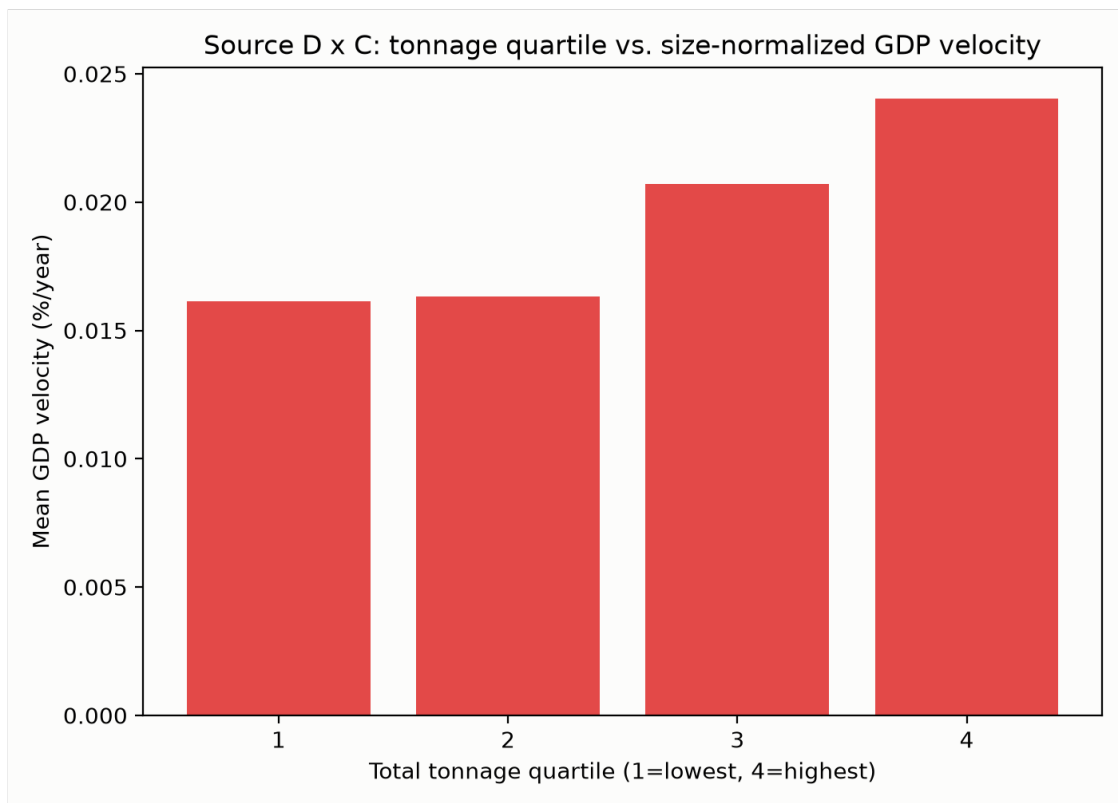


Figure 3. Mean size-normalized GDP velocity by tonnage quartile. Shows the clean monotonic step described above, with Q4 roughly 1.5x Q1.

All correlations are weak in absolute terms, consistent with the pattern Source F's cross-validation against Source C already established (structural/static signals track short-run momentum only faintly). But unlike Source F's near-zero, directionless result, this one is monotonic: each tonnage quartile step up trends toward faster GDP growth, a small but economically sensible signal that higher freight throughput associates with (not necessarily causes) faster local growth.

3.5 Limitations / Open Items

- **No dollar-value column** — see §4. Any downstream use expecting a value-weighted (not just tonnage-weighted) trade signal is not supported by this experimental county-level product.
- **Single 2022 cross-section, not a time series** — unlike Source C's rolling velocity window, Source D captures one snapshot; trend/momentum in freight flows themselves cannot be derived from this data alone.
- **dms_mode code 11 remains unexplained** — the experimental county product's mode taxonomy doesn't match the general FAF5 user guide's 1–8 code table; the county-level-specific technical report that would likely resolve this sits behind the same Akamai gate as the landing page and wasn't pursued.
- **Zone → county crosswalk was not pursued** — HHI pooling uses raw FAF-zone partner rows without resolving them to zone-centroid geography, so no distance/reach metric was salvaged; recommended if a "reach" signal is wanted in a future round.
- **Ingestion has no per-state checkpointing** — a known, accepted limitation for the current run (which completed in 2h32min with zero failures); revisit before assuming a future data refresh will be equally fast.

4. Proposal Alignment Assessment

The proposal frames Source D as isolating "a logistical pass-through corridor or industrial exporter from a pure consumer sink" via county-to-county freight flows, and the pre-scoping spec specifically worried about a "combinatorial matrix explosion" from a naive dense county×county matrix, proposing top-K-per-origin truncation as a fix while flagging that truncation "destroys global network topology."

- **Supported:** the core framing holds at national scale (§3.2, §3.3) — scalar tonnage cleanly separates known hubs (Houston, LA, Chicago) from rural sinks by two orders of magnitude, and partner concentration adds a second, complementary axis (hubs funnel volume through fewer dominant corridors) once tested at full national scale rather than a two-state regional sample.
- **Deviation 1 — combinatorial-explosion problem already solved upstream, differently than proposed:** BTS's own county/FAF-zone hybrid table structure (full county granularity only for nearby geography, FAF-zone aggregation for everything farther) already prevents the dense 9.87M-edge matrix both scoping docs worried about. This supersedes the spec's proposed top-K-per-origin truncation entirely — that scheme would have solved a problem the source data doesn't actually present in that form, and would have discarded the zone-level long-tail structure BTS already preserves.
- **Deviation 2 — no dollar-value column:** both scoping docs describe Source D as tracking "freight tonnage and dollar value." The experimental county-level product ships tonnage only (`tons_2022`); the base (non-experimental) FAF5 product does carry `value/current_value` fields, per the general user guide, but that field was dropped in the county-level disaggregation.

This is a real scope reduction, not a parsing gap — confirmed against the FAF5 data dictionary directly.

- **New finding not anticipated by either doc:** the HHI concentration signal's *direction* is scale-dependent (§3.3) — a design detail neither scoping doc could have anticipated, since neither analyzed the disaggregation model's gravity-artifact behavior at regional vs. national scope. Anyone extending this concentration metric should be aware it only reads correctly at full national coverage.

5. Next Actions

- **Test the proposal's stated Trade Logistics synergy with Source F**, which explicitly flagged this as its own next action: does Source F's structural typology (industry dependence, demographic distress) explain which counties become logistics hubs vs. sinks, beyond what tonnage/HHI alone show?
- **Once Source B (BLS QCEW) exists**, test whether a county's industry mix explains its trade-flow character — the proposal's stated three-way synergy between Movement, Capital Flow, and industry composition.
- No action planned this round on the `dms_mode=11` documentation gap (§3.5) or the zone-centroid crosswalk (§3.5) — revisit only if a downstream consumer needs either resolved.

Appendix — Key Statistics

Metric	Value
Total counties	3,144
Top hub county (by tonnage)	Harris County, TX (731,121 tons)
log10(tons) vs. mean partner HHI (Pearson r)	0.2784
log(tons) vs. unemployment velocity (Pearson r, permutation p)	0.0498, 0.0060
log(tons) vs. size-normalized GDP velocity (Pearson r, permutation p)	0.0765, 0.0020
Partner HHI vs. unemployment velocity (Pearson r, permutation p)	-0.0261, 0.164
Partner HHI vs. size-normalized GDP velocity (Pearson r, permutation p)	0.0688, 0.0020
GDP velocity %, tonnage quartile 1 (lowest)	1.61%
GDP velocity %, tonnage quartile 2	1.63%

GDP velocity %, tonnage quartile 3	2.07%
GDP velocity %, tonnage quartile 4 (highest)	2.40%

Source: analysis-output/source_d_stats.json, source-d-findings.md, figures/source-d-numeric-summary.md (MacroEmbeddings repo).